

Radiative transitions of B_{c1} states in hard-photon QCD sum rules

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Abstract

We study the radiative transitions $B_{c1}(6743) \rightarrow B_c \gamma$, $B_{c1}(6743) \rightarrow B_c^* \gamma$, $B_{c1}(6750) \rightarrow B_c \gamma$, and $B_{c1}(6750) \rightarrow B_c^* \gamma$ in a heavy-heavy three-point QCD sum-rule framework. Since both valence quarks are heavy, the leading radiative contribution is hard photon emission from the charm and anti-bottom lines rather than a photon-distribution-amplitude term. Using physical-current residues from the two-point B_c -mixing sum rules, we find

$$\Gamma[B_{c1}(6743) \rightarrow B_c \gamma] = 0.532^{+0.202}_{-0.146} \text{ keV}, \quad \Gamma[B_{c1}(6750) \rightarrow B_c \gamma] = 10.04^{+2.41}_{-1.75} \text{ keV}.$$

For the vector final state, using the standard vector-current normalization $f_{B_c^*} = 0.435^{+0.032}_{-0.035} \text{ GeV}$, the present current convention gives

$$\Gamma[B_{c1}(6743) \rightarrow B_c^* \gamma] = 70.3^{+10.9}_{-8.4} \text{ keV}, \quad \Gamma[B_{c1}(6750) \rightarrow B_c^* \gamma] = 382^{+55}_{-50} \text{ keV}.$$

These values include the explicitly integrated all-positive dimension-four gluon-condensate sector. The contact/derivative rows have no ordinary two-channel double-pole support in the physical double-Borel projection. A separate sign and state-assignment audit shows that the vector hierarchy is opposite to representative model calculations. An independent FeynCalc commutator check confirms the tensor-current sign used here, so the reversal is traced to the physical-current mixing interference in the adopted state assignment.

I. INTRODUCTION

The spectroscopy of the B_c family is a particularly clean laboratory for heavy-heavy dynamics. Unlike ordinary heavy-light mesons, the B_c mesons contain two different heavy flavors, and their electromagnetic transitions are sensitive to the relative photon couplings of the charm and anti-bottom quark. Radiative transitions of orbitally excited B_c states therefore provide a useful complement to mass spectroscopy and hadronic transitions.

The observation of structures in the $B_c^+ \gamma$ spectrum by LHCb [1, 2] motivates a direct QCD sum-rule calculation of the corresponding radiative widths. The experimental analysis observes peaks in the visible $B_c^+ \gamma$ invariant mass, but does not measure the individual partial radiative widths. Theoretical predictions for $B_{c1} \rightarrow B_c^{(*)} \gamma$ have been obtained in several potential model and relativistic quark-model approaches [3–11]. The spread among predictions is sizeable, especially for channels in which the spin components of the axial state interfere.

In the strange heavy systems, light-cone QCD sum rules receive important photon-distribution-amplitude contributions. The B_c case is different: there is no light valence quark, and the leading radiative term is the short-distance photon emission from heavy propagators. In this paper we build a hard-photon three-point QCD sum rule for the transitions $B_{c1} \rightarrow B_c \gamma$ and $B_{c1} \rightarrow B_c^* \gamma$. The axial-vector currents and residues are matched to the same physical-current basis used in the B_c -mixing two-point analysis. We also give a detailed audit of the dimension-four gluon-condensate sector, since in a heavy-heavy system it is the first natural nonperturbative correction to the perturbative triangle.

The paper is organized as follows. In Sec. II we introduce the interpolating currents, the physical mixing convention, and the radiative form factors. Section III summarizes the hard-photon sum-rule construction and the condensate audit. Numerical inputs, stability checks, and comparison with experiment and other theory are given in Sec. IV. Section V summarizes the results.

II. THEORETICAL FRAMEWORK

We use two independent axial-vector basis currents,

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad (1)$$

$$J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} P^\alpha \gamma_5 c, \quad (2)$$

where P is the momentum of the initial axial-vector state. The physical currents are written as

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad (3)$$

$$J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B. \quad (4)$$

The lower state $B_{c1}(6743)$ is coupled to $J_{1\mu}$, while the upper state $B_{c1}(6750)$ is coupled to $J_{2\mu}$. The two-point sum-rule diagonalization gives $\theta \simeq 43.3^\circ$ in the numerical window used below.

The decay constants are defined by

$$\langle 0 | \bar{b} i \gamma_5 c | B_c(p) \rangle = \frac{m_{B_c}^2 f_{B_c}}{m_b + m_c}, \quad (5)$$

$$\langle 0 | \bar{b} \gamma_\rho c | B_c^*(p, \xi) \rangle = m_{B_c^*} f_{B_c^*} \xi_\rho, \quad (6)$$

$$\langle 0 | J_{i\mu} | B_{c1}^{(i)}(P, \eta) \rangle = f_i m_i \eta_\mu, \quad i = 1, 2. \quad (7)$$

For the pseudoscalar final state we parametrize the radiative matrix element as

$$\langle B_c(p)\gamma(q,\varepsilon)|B_{c1}^{(i)}(P,\eta)\rangle = eg_i [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)]. \quad (8)$$

The corresponding width is

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} g_i^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i} \right)^3. \quad (9)$$

For the vector final state we use the gauge-invariant tensor

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho\alpha\beta} p^\alpha q^\beta - (p \cdot q) \epsilon_{\mu\nu\rho\alpha} p^\alpha, \quad (10)$$

and define

$$\langle B_c^*(p,\xi)\gamma(q,\varepsilon)|B_{c1}^{(i)}(P,\eta)\rangle = eg_i^* \eta^\mu \varepsilon^{*\nu} \xi^{*\rho} V_{\mu\nu\rho}. \quad (11)$$

The width is evaluated by an explicit polarization sum.

III. QCD SUM RULES

For the $B_c\gamma$ channel the correlation function is

$$\Pi_{i\mu}(p,q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q,\varepsilon) | T \{ J_{i\mu}(x) \bar{c}(0) i\gamma_5 b(0) \} | 0 \rangle. \quad (12)$$

For the $B_c^*\gamma$ channel the pseudoscalar current is replaced by $\bar{c}\gamma_\rho b$. Since both quarks are heavy, the leading OPE is obtained by attaching the photon perturbatively to either the charm line or the anti-bottom line. In the charm-emission topology the three denominators are

$$D_1 = (k+q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k-p)^2 - m_b^2, \quad (13)$$

while in the anti-bottom-emission topology they are

$$D_1 = k^2 - m_c^2, \quad D_2 = (k-p)^2 - m_b^2, \quad D_3 = (k-p+q)^2 - m_b^2. \quad (14)$$

The electric charges are $e_c = 2/3$ and $e_{\bar{b}} = 1/3$. After the Dirac trace and projection, both emission topologies can be written through the kernel

$$\mathcal{R}(s; m_i, m_j, a, b) = -\frac{3}{8\pi^2} \left[2a \ln \frac{s - m_j^2 + m_i^2 - \lambda(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda(s, m_j^2, m_i^2)} + \frac{b}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \lambda(s, m_j^2, m_i^2) \right], \quad (15)$$

TABLE I. Coefficients entering Eq. (16). The labels P and V denote the pseudoscalar and vector final state, respectively.

Density	a_c^X	b_c^X	a_b^X	b_b^X
P, A	m_c	$m_c^2(m_c - m_b)$	m_b	$m_b^2(m_c - m_b)$
P, B	$(2m_c^2 - m_b m_c)/(m_b + m_c)$	$m_c^2 s/(m_b + m_c)$	$m_b m_c/(m_b + m_c)$	$m_b^2 s/(m_b + m_c)$
V, A	$(-4m_b m_c + 2m_c^2)/s$	$4m_c^2$	$(-2m_b^2 + 4m_b m_c)/s$	$4m_b^2$
V, B	$\frac{2m_c}{m_b + m_c} + \frac{2m_b^2 m_c - 4m_b m_c^2 + 2m_c^3 + 2m_c p \cdot q}{(m_b + m_c)s}$	$\frac{-4m_b m_c^2 + 4m_c^3}{m_b + m_c}$	$\frac{2m_b}{m_b + m_c} + \frac{-6m_b^3 + 4m_b^2 m_c + 2m_b m_c^2 + 2m_b p \cdot q}{(m_b + m_c)s}$	$\frac{-4m_b^3 + 4m_b^2 m_c}{m_b + m_c}$

where

$$\lambda(s, m_j^2, m_i^2) = \sqrt{[s - (m_i + m_j)^2][s - (m_i - m_j)^2]}.$$

The perturbative densities are

$$\rho_X(s) = e_{\bar{b}} \mathcal{R}(s; m_b, m_c, a_b^X, b_b^X) - e_c \mathcal{R}(s; m_c, m_b, a_c^X, b_c^X), \quad (16)$$

with the coefficients shown in Table I. For the vector J_B density the scalar $p \cdot q$ is evaluated at the physical value $(m_i^2 - m_{B_c^*}^2)/2$ of the channel.

The Borel-projected amplitudes are then

$$\widehat{\Pi}_X(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds e^{-s/M^2} \rho_X(s) + \widehat{\Pi}_X^{G^2}.$$

The physical amplitudes are obtained by rotating the basis amplitudes,

$$\widehat{\Pi}_1 = \sin \theta \widehat{\Pi}_A + \cos \theta \widehat{\Pi}_B, \quad (17)$$

$$\widehat{\Pi}_2 = \cos \theta \widehat{\Pi}_A - \sin \theta \widehat{\Pi}_B. \quad (18)$$

The resulting form factors have the schematic form

$$g_1 = \frac{\exp[(m_1^2 + m_f^2)/(2M^2)]}{m_1 f_1 \mathcal{N}_f} \left(\sin \theta \widehat{\Pi}_A + \cos \theta \widehat{\Pi}_B \right), \quad (19)$$

$$g_2 = \frac{\exp[(m_2^2 + m_f^2)/(2M^2)]}{m_2 f_2 \mathcal{N}_f} \left(\cos \theta \widehat{\Pi}_A - \sin \theta \widehat{\Pi}_B \right), \quad (20)$$

where $\mathcal{N}_f = m_{B_c}^2 f_{B_c}/(m_b + m_c)$ for $B_c \gamma$ and $\mathcal{N}_f = m_{B_c^*}^2 f_{B_c^*}$ for $B_c^* \gamma$. For the vector final state we extract g_V with

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho p q} - (p \cdot q) \epsilon_{\mu\nu\rho p}.$$

Since this tensor has mass dimension three, g_V has dimension GeV^{-2} . The polarization sum gives

$$\Gamma(1^+ \rightarrow 1^- \gamma) = \frac{\alpha_{\text{em}}}{3} g_V^2 (2m_f^2 + E_\gamma^2) E_\gamma^3,$$

Equivalently,

$$g_{E1} = \sqrt{2m_f^2 + E_\gamma^2} g_V, \quad \Gamma = \frac{\alpha_{\text{em}}}{3} g_{E1}^2 E_\gamma^3.$$

The dimension-four gluon-condensate workbench uses the background-field heavy propagator expansion. The projected G^2 numerators were reduced in powers of D_1, D_2, D_3 . For the vector channel this gives 1095 reduced terms: 538 with all effective denominator powers positive and 557 contact/derivative-sector terms. For the pseudoscalar channel the corresponding all-positive sector contains 375 terms. We integrate these all-positive terms explicitly in the diagonal double-Borel limit $M_1^2 = M_2^2 = 2M^2$. Refined support audits show that the contact/derivative rows do not contain ordinary two-channel double poles. The vector crossed rows depend on $(p - q)^2 = 2p^2 - P^2$, and the pseudoscalar contact rows are either single-virtuality, no-virtuality, or crossed-combination terms. They are kept as audit candidates rather than added to the physical double-Borel sum.

IV. NUMERICAL ANALYSIS

We use $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, $m_{B_c} = 6.2749$ GeV, and $m_{B_c^*} = 6.3359$ GeV. The axial-state masses are $m_1 = 6.743$ GeV and $m_2 = 6.751$ GeV. The working Borel and continuum windows are

$$M^2 = 7, 8, 9 \text{ GeV}^2, \quad s_0 = 53, 54, 55 \text{ GeV}^2,$$

and the vector-current two-point sum rule for $f_{B_c^*}$ uses

$$M_V^2 = 5, 6, 7, 8 \text{ GeV}^2, \quad s_{0,V} = 40, 42, 44 \text{ GeV}^2.$$

The two-point residue grid gives

$$\theta = 43.29^\circ, \quad f_1 = 0.264[0.259, 0.269] \text{ GeV}, \quad f_2 = 0.722[0.709, 0.738] \text{ GeV}.$$

For the vector final-state decay constant we find

$$f_{B_c^*} = 0.435[0.400, 0.467] \text{ GeV}$$

in the standard vector-invariant convention.

The final uncertainties are obtained by sampling $m_b, m_c, \theta, f_{B_c}$, the Borel parameter, and the continuum threshold over the stated windows, and then folding in the explicitly

TABLE II. Numerical inputs and sum-rule windows used in the analysis. The quoted residue and $f_{B_c^*}$ intervals are the 16–84 percentile windows from the corresponding two-point sum-rule grids.

Quantity	Value or window
m_b, m_c	4.18 GeV, 1.27 GeV
$m_{B_c}, m_{B_c^*}$	6.2749 GeV, 6.3359 GeV
m_1, m_2	6.743 GeV, 6.751 GeV
f_{B_c}	0.371 ± 0.037 GeV
$f_{B_c^*}$	0.435[0.400, 0.467] GeV
θ	$43.29^\circ[43.10^\circ, 43.44^\circ]$
f_1, f_2	0.264[0.259, 0.269] GeV, 0.722[0.709, 0.738] GeV
Radiative M^2, s_0	7–9 GeV ² , 53–55 GeV ²
Vector-current $M_V^2, s_{0,V}$	5–8 GeV ² , 40–44 GeV ²

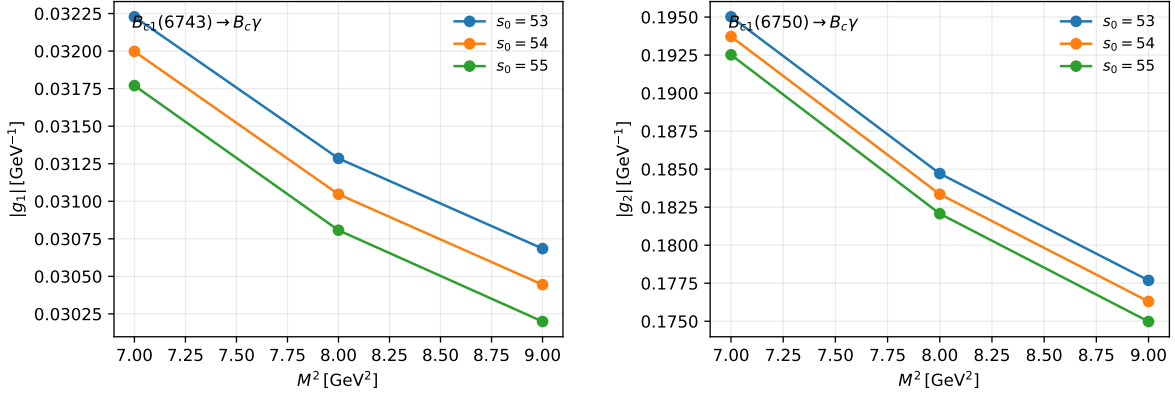


FIG. 1. Representative Borel stability of the pseudoscalar final-state form factors.

integrated all-positive G^2 contribution. We quote the median and the 16–84 percentile interval rather than a Gaussian fit, since the mixed amplitudes are interference sensitive.

Table IV compares our results with representative model predictions. LHCb has observed the $B_c^+ \gamma$ structure, but the individual partial radiative widths are not measured [1, 2]; hence the comparison is with theory rather than direct width measurements.

† Convention-fixed mixed-current QCDSR values in the adopted two-point state assignment; their hierarchy differs from representative model calculations because of the mixing interference discussed below.

TABLE III. Present hard-QCDSR results. The quoted intervals are the 16–84 percentile Monte Carlo intervals after including the explicitly integrated all-positive G^2 sector. The $B_c^*\gamma$ entries use the tensor-current sign fixed by the independent commutator audit discussed in the text.

Channel	Coupling	Γ [keV]
$B_{c1}(6743) \rightarrow B_c\gamma$	$g_P = +0.0487[0.0414, 0.0572] \text{ GeV}^{-1}$	$0.532^{+0.202}_{-0.146}$
$B_{c1}(6750) \rightarrow B_c\gamma$	$g_P = -0.206[-0.230, -0.188] \text{ GeV}^{-1}$	$10.04^{+2.41}_{-1.75}$
$B_{c1}(6743) \rightarrow B_c^*\gamma$	$g_V = -0.0764[-0.0821, -0.0717] \text{ GeV}^{-2}$	$70.3^{+10.9}_{-8.4}$
$B_{c1}(6750) \rightarrow B_c^*\gamma$	$g_V = +0.173[0.161, 0.185] \text{ GeV}^{-2}$	382^{+55}_{-50}

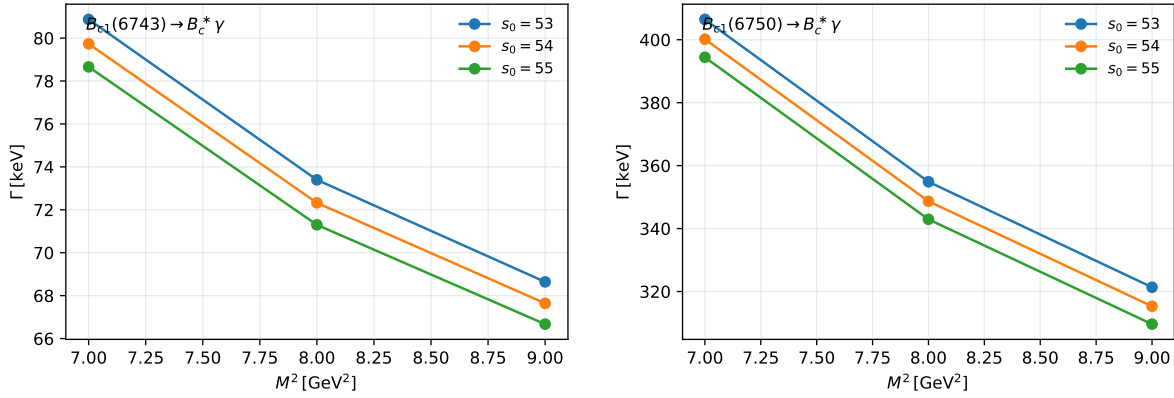


FIG. 2. Representative Borel stability of the vector final-state widths in the standard $f_{B_c^*}$ normalization.

The small $B_{c1}(6743) \rightarrow B_c\gamma$ width and the comparatively large $B_{c1}(6750) \rightarrow B_c^*\gamma$ width reflect the strong dependence of the radiative amplitude on the physical-current rotation and on the final-state spin. The vector channels are also sensitive to the normalization of $f_{B_c^*}$; using the reference value $f_{B_c^*} = 0.384 \pm 0.038 \text{ GeV}$ instead of the self-contained standard sum-rule value gives $\Gamma(6743 \rightarrow B_c^*\gamma) = 92.6[76.1, 117] \text{ keV}$ and $\Gamma(6750 \rightarrow B_c^*\gamma) = 447[365, 564] \text{ keV}$. More importantly, the physical mixed-current vector ratio $\Gamma(6750 \rightarrow B_c^*\gamma)/\Gamma(6743 \rightarrow B_c^*\gamma) > 1$ is opposite to the representative model pattern. At the central point the lower-state vector combination is cancellation-dominated, $\sin\theta \hat{\Pi}_A + \cos\theta \hat{\Pi}_B \simeq -1.8 \times 10^{-3}$, whereas the higher-state combination is constructive, $\cos\theta \hat{\Pi}_A - \sin\theta \hat{\Pi}_B \simeq 1.05 \times 10^{-2}$. As a diagnostic, the unmixed pure-basis limits do not show this reversal: pure J_A gives 878 and 126 keV for the 6743 and 6750 states, respectively, while pure J_B gives 1554

TABLE IV. Comparison with representative theoretical predictions. Widths are in keV.

Publication	$6743 \rightarrow B_c \gamma$	$6743 \rightarrow B_c^* \gamma$	$6750 \rightarrow B_c \gamma$	$6750 \rightarrow B_c^* \gamma$
Godfrey [3]	13	60	80	11
Ebert–Faustov–Galkin [4]	18.4	78.9	132	13.6
Fulcher [5]	32.4	75.6	127.8	26.2
Gershtein et al. [6]	11.6	77.8	131.1	8.1
T. Y. Li et al. [7]	16.6	49	66.6	10.5
Q. Li et al. [8]	35	70	74	40
X. J. Li et al. [9]	30.1	47.8	64	25.6
Eichten–Quigg [10]	9.9	62.5	92.3	7.5
Bondar–Milstein [11]	22	75	47	2
This work	0.532	70.3 [†]	10.04	382 [†]

TABLE V. Central-point diagnostic for the $B_c^* \gamma$ hierarchy at $M^2 = 8 \text{ GeV}^2$, $s_0 = 54 \text{ GeV}^2$. The pure-basis rows are diagnostic checks only; the physical rows are obtained with the mixed currents in Eq. (4).

Case	$\Gamma_{6743} [\text{keV}]$	$\Gamma_{6750} [\text{keV}]$	high/low
pure J_A	878	126	0.143
pure J_B	1554	222	0.143
physical mixing, perturbative	70.2	338	4.82
physical mixing, with G_{ap}^2	70.3	382	5.43

and 222 keV. The no-mixing ratios are therefore below unity, in qualitative agreement with the model hierarchy; the opposite baseline mixed-current ratio is caused by the interference pattern above. A flip of the relative J_B radiative sign would restore the model-like hierarchy, but this sign flip is not supported by the tensor-current definition used in the two-point mixing analysis. We checked this explicitly by replacing $i\sigma_{\mu\alpha}P^\alpha\gamma_5$ with the commutator form $-\gamma_\mu\gamma_\alpha P^\alpha\gamma_5/2$, which gives the same projected J_B amplitude. Reversing the current phase or the tensor-index order flips the amplitude, but that is a different current convention. Within the adopted convention, the vector hierarchy is therefore a physical mixed-current interference effect.

V. CONCLUSION

We have calculated the leading hard-photon QCD sum-rule baselines for $B_{c1} \rightarrow B_c \gamma$ and $B_{c1} \rightarrow B_c^* \gamma$ transitions. The calculation uses physical axial-current residues from the two-point B_c -mixing sum rules and a self-contained vector-current sum rule for $f_{B_c^*}$. The resulting widths show a strong channel hierarchy: $B_{c1}(6743) \rightarrow B_c \gamma$ is very small, $B_{c1}(6750) \rightarrow B_c \gamma$ is at the few-keV level. The vector final-state calculation is internally normalized, and the independent commutator audit fixes the relative tensor-current sign used in the J_B amplitude.

The dimension-four gluon-condensate sector has been reduced algebraically, audited, and included through the all-positive denominator sector. The contact/derivative rows do not have ordinary two-channel double-pole support in the physical double-Borel projection, so they are not added to the sum rule. The present calculation therefore gives a transparent QCDSR prediction for $B_c \gamma$ and a convention-fixed QCDSR prediction for $B_c^* \gamma$ in the adopted two-point state assignment. The contrast with model hierarchies is not a phase-space effect; it is caused by destructive interference in the 6743 mixed-current amplitude and constructive interference in the 6750 amplitude.

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